



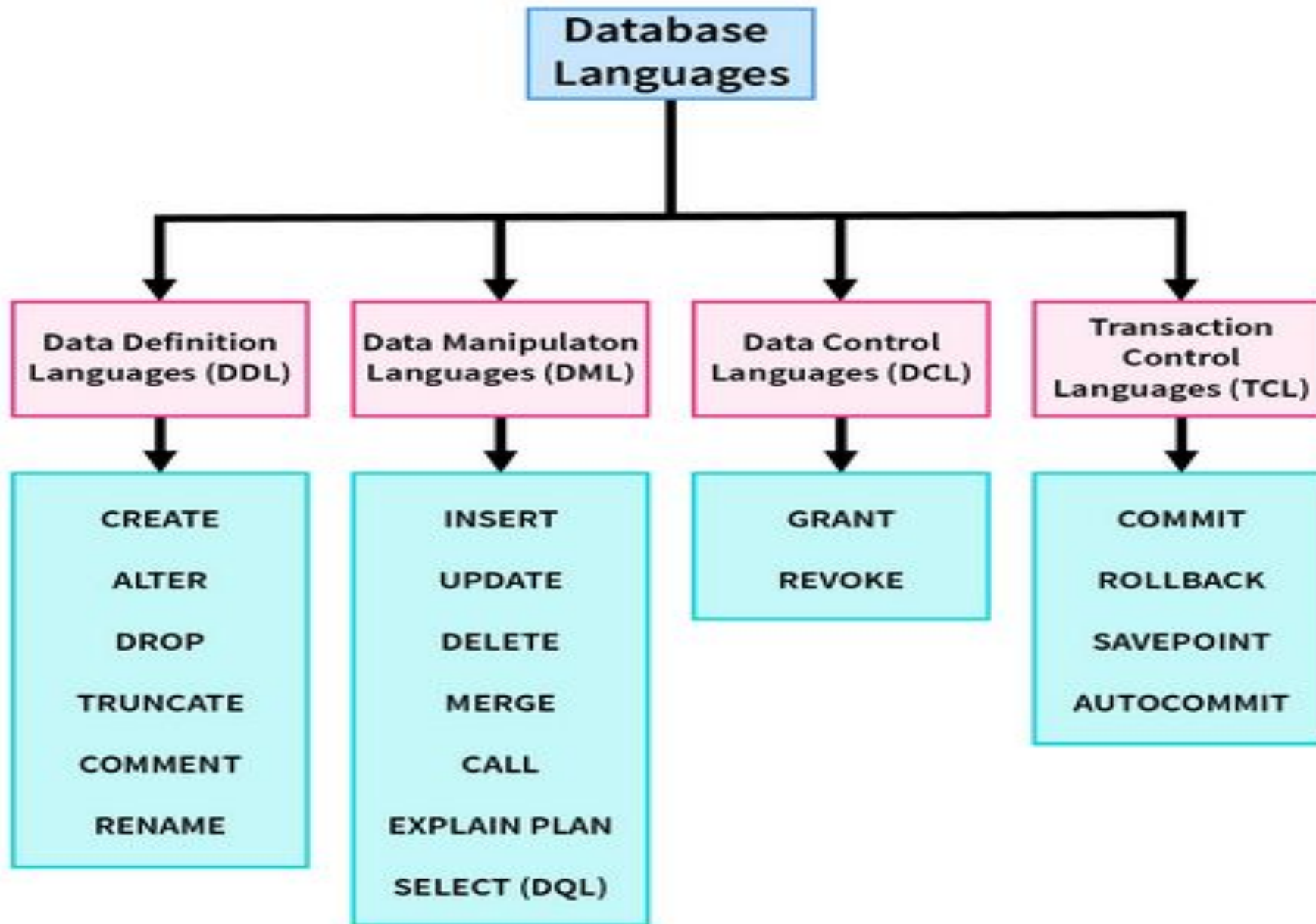
Database Languages in DBMS

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Database Language

- ▶ Database Language is a particular type of programming language used to define and manipulate a database.
- ▶ Database languages are used to perform various critical tasks that help a database management system function correctly.
- ▶ A DBMS has appropriate languages and interfaces to express database queries and updates.
- ▶ Database languages can be used to read, store and update the data in the database.
- ▶ Based on their application, database languages are classified into four types: DDL, DML, DCL, and TCL.

Types of Database Languages



Data Definition Language (DDL)

- ▶ **DDL** stands for **D**ata **D**efinition **L**anguage. It is used to define database structure or pattern.
- ▶ It is used to create schema, tables, indexes, constraints, etc. in the database.
- ▶ Using the DDL statements, you can create the skeleton of the database.
- ▶ Data definition language is used to store the information of metadata like the number of tables and schemas, their names, indexes, columns in each table, constraints, etc.

Here are some tasks that come under DDL:

- ▶ **Create:** It is used to create objects in the database.
- ▶ **Alter:** It is used to alter the structure of the database.
- ▶ **Drop:** It is used to delete objects from the database.
- ▶ **Truncate:** It is used to remove all records from a table.
- ▶ **Rename:** It is used to rename an object.
- ▶ **Comment:** It is used to comment on the data dictionary.
- ▶ These commands are used to update the database schema that's why they come under Data definition language.

Data Manipulation Language (DML)

- ▶ **DML** stands for **Data Manipulation Language**. It is used for accessing and manipulating data in a database. It handles user requests.
- ▶ Here are some tasks that come under DML:
- ▶ **Select**: It is used to retrieve data from a database.
- ▶ **Insert**: It is used to insert data into a table.
- ▶ **Update**: It is used to update existing data within a table.
- ▶ **Delete**: It is used to delete all records from a table.
- ▶ **Merge**: It performs UPSERT operation, i.e., insert or update operations.
- ▶ **Call**: It is used to call a structured query language or a Java subprogram.
- ▶ **Explain Plan**: It has the parameter of explaining data.
- ▶ **Lock Table**: It controls concurrency.

Data Control Language (DCL)

- ▶ **DCL** stands for **D**ata **C**ontrol **L**anguage. It is used to retrieve the stored or saved data.
- ▶ The DCL execution is transactional. It also has rollback parameters.
- ▶ Here are some tasks that come under DCL:
- ▶ **Grant:** It is used to give user access privileges to a database.
- ▶ **Revoke:** It is used to take back permissions from the user.
- ▶ There are the following operations which have the authorization of Revoke:
- ▶ CONNECT, INSERT, USAGE, EXECUTE, DELETE, UPDATE and SELECT.

Transaction Control Language (TCL)

- ▶ TCL is used to run the changes made by the DML statement. TCL can be grouped into a logical transaction.
- ▶ Here are some tasks that come under TCL:
- ▶ **Commit:** It is used to save the transaction on the database.
- ▶ **Rollback:** It is used to restore the database to original since the last Commit.

What is Structured Query Language (SQL)?

- ▶ Structured Query Language (SQL) is a standardized programming language that is used to manage relational databases and perform various operations on the data in them. Initially created in the 1970s, SQL is regularly used not only by database administrators but also by developers writing data integration scripts and data analysts looking to set up and run analytical queries.
- ▶ The term SQL is pronounced *ess-kew-ell* or *sequel*.

SQL is used for the following:

- ▶ Modifying database table and index structures.
- ▶ Adding, updating and deleting rows of data.
- ▶ Retrieving subsets of information from within relational database management systems ([RDBMSes](#)). This information can be used for transaction processing, analytics applications and other applications that require communicating with a relational database



- ▶ A table is the most basic unit of a database and consists of rows and columns of data. A single table holds records, and each record is stored in a row of the table. Tables are the most used type of database objects or structures that hold or reference data in a relational database. Other types of database objects include the following:
- ▶ **Views** are logical representations of data assembled from one or more database tables.
- ▶ **Indexes** are lookup tables that help speed up database lookup functions.
- ▶ **Reports** consist of data retrieved from one or more tables, usually a subset of that data that is selected based on search criteria.